# EXPERIMENT 18

## Implement image cropping, copying and pasting to select a region of interest (ROI).

### Aim

Implement image cropping, copying and pasting to select a region of interest (ROI).

### Algorithm

• Import OpenCV and read the image.

• Select a rectangular region of interest.

• Copy the ROI to a separate array.

• Paste the ROI at another location in a copy of the image.

• Save the cropped and pasted images.

### Code

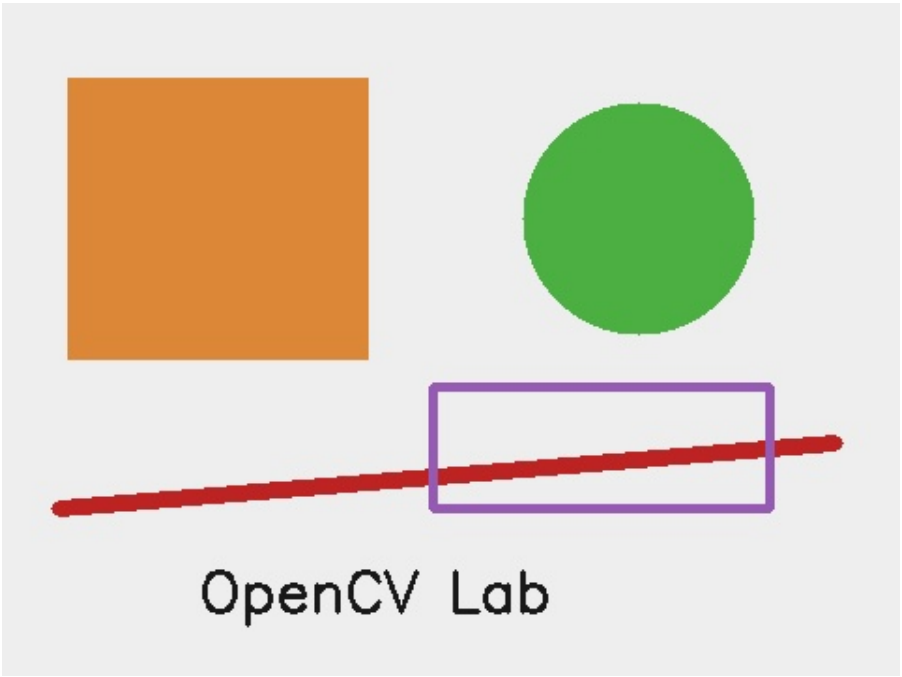
```
import cv2

img = cv2.imread("images/input.jpg")
if img is None:
    raise FileNotFoundError("images/input.jpg not found")
h, w = img.shape[:2]
crop = img[h // 4:h // 2, w // 4:w // 2].copy()
ch, cw = crop.shape[:2]
result = img.copy()
result[0:ch, 0:cw] = crop
cv2.imwrite("output_images/experiment18_roi_crop.png", crop)
cv2.imwrite("output_images/experiment18_roi_pasted.png", result)
print("Saved cropped and pasted images")
```

### Input Image:

### Output Image:

### Input Image




### Output Image